

Standard Operating Procedure

Access Control Power Supply Sizing

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Related Documents	SOP-LV-015, SOP-LV-014
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Owner	Apex – Operations
Approved By	_____

NOTE ON THIS DOCUMENT:

This SOP covers general power supply sizing methodology for access control systems using UL294-listed equipment. Required battery backup runtime is an application/AHJ-driven decision, not a fixed number this SOP can specify — confirm the required backup duration for each specific site before sizing batteries, rather than assuming a default.

1. Purpose

This procedure establishes the standard method for calculating access control power supply and battery backup capacity requirements, to prevent undersized power supplies that cause intermittent lock failures, voltage sag under load, or inadequate backup runtime.

2. Scope

Applies to Apex technicians and estimators sizing power supplies for Kantech controllers, locks, readers, and associated access control devices.

3. UL294 — What It Requires

UL 294 is the governing UL standard for access control system power supplies. A UL294-listed power supply is tested for filtered, regulated output, extended full-load duration, electric shock resistance, and (where equipped) battery backup/charging performance. Only use UL294-listed power supplies for access control applications — do not substitute a generic unlisted DC supply, even if the voltage and current rating appear to match on paper.

- UL294 power supplies are rated at +10% of stated voltage — account for this when calculating voltage drop margins, not just the nominal rated voltage.
- Confirm AC fail supervision and low-battery supervision outputs are wired back to the access control system or a monitored point where the application requires supervised power status — these supervisory outputs exist on most UL294 supplies specifically to be used, not left disconnected.

4. Calculating Continuous Load

Sum the continuous (always-on) current draw of every device the supply will power:

- Controllers (idle current draw, per the specific model's datasheet)
- Readers (each reader's rated current draw — confirm per model, do not assume all readers draw the same current)
- Any continuously-energized locks (fail-safe strikes or maglocks held locked/energized continuously, depending on application)
- Auxiliary devices continuously powered from the same supply (REX sensors, door contacts with LED indicators, etc.)

5. Calculating Intermittent / Peak Load

Separately account for loads that only draw current briefly:

- Lock strike current during the momentary unlock pulse (higher inrush than steady-state in some strike designs — confirm against the specific strike's datasheet)
- Door operator or auxiliary relay activation current, where applicable
- Multiple doors unlocking simultaneously (e.g., a fire alarm release event, or a scheduled unlock across several doors) — size for the realistic worst case of simultaneous activation on that supply, not just one door at a time, if the application could actually trigger that.

6. Sizing the Power Supply

- Total the continuous load and add a safety margin — do not size a supply to its rated maximum with zero headroom; running a supply at its absolute limit shortens its service life and leaves no margin for future device additions.
- Confirm the supply's rated output current comfortably exceeds the calculated continuous load, and separately confirm it can handle the peak/intermittent load without unacceptable voltage sag, per the manufacturer's load-regulation specification.
- Where a single supply would be marginal or exceed one enclosure's practical output, use multiple supplies/enclosures rather than overloading one unit — confirm with the Site Supervisor at the design stage, not after installation has begun.

7. Battery Backup Sizing

Where battery backup is required (confirm the required runtime for the specific site/application — this varies and is not a fixed universal number):

- Required battery capacity (Ah) is approximately: total DC load current (A) × required backup duration (hours), with a safety margin added — batteries lose usable capacity as they age and are rarely used at their full rated Ah in practice.
- Confirm the power supply's built-in charger can fully recharge the selected battery within a reasonable window — an oversized battery paired with an undersized charger may never reach full charge between outages.
- Confirm the physical battery fits the enclosure — enclosures are typically rated for a specific number and size of batteries; do not assume a larger battery will fit without checking.
- Test actual backup runtime periodically rather than relying solely on the calculated value — real-world battery performance degrades over time and calculated capacity is a starting point, not a guarantee.

8. Documentation

Record on the site power/load schedule: total continuous load, peak load, power supply model and rated output, battery model and Ah rating, and calculated backup runtime. This becomes part of the site's commissioning file and is the reference point for any future load additions to that supply.

9. References

- UL 294 — Access Control System Units
- Manufacturer datasheets for the specific power supply, controllers, locks, and readers in use (govern where more restrictive than this SOP)
- SOP-LV-015 — Electric Strike & Maglock Installation
- SOP-LV-014 — Fire Alarm Interface & Life-Safety Interlock